

- 2 (a) (i) gravitational field strength is equal to the negative gravitational potential gradient B1
 Or state equation $g = - d\phi/dr$
- (b) (i) $G = GM/r^2$ C1
 $g_A = G(3M)/(2R)^2 = \frac{3}{4} GM/R^2$
 $g_B = G(M)/(R)^2 = GM/R^2$
 so conclude $g_A < g_B$ A1
- alternative soln:
 by calculating the gradient at surface and using gradient = field strength
 deduce gradient at surface of A < gradient at surface of B
- (ii) M is the point/position in which net gravitational field strength is zero B1
- (iii) sketch: one curve, starting with gradient of decreasing magnitude at 2R B1
 and finishing with gradient of increasing magnitude at D – R
 field strength shown as zero near the point of maximum potential B1
 (position of M correct), with gravitation field strength at surface of A < B

- (c) (i) The gravitational force of attraction between the stars are of the same magnitude. This force provides the centripetal force for each star to orbit about their common centre of mass. B1
- (ii) There is no external force acting on the binary stars, so the c.m. should not change position. B1
- (iii) centripetal force = $M_A d \omega^2 = M_B (2.8 \times 10^8 - d) \omega^2$ C1
 Or $M_A d_A = M_B d_B$
 $M_A / M_B = 3.0 = (2.8 \times 10^8 - d) / d$
 $d = 7.0 \times 10^7 \text{ km}$ A1